DEEP LEARNING-BASED THERMAL FAULT DETECTION

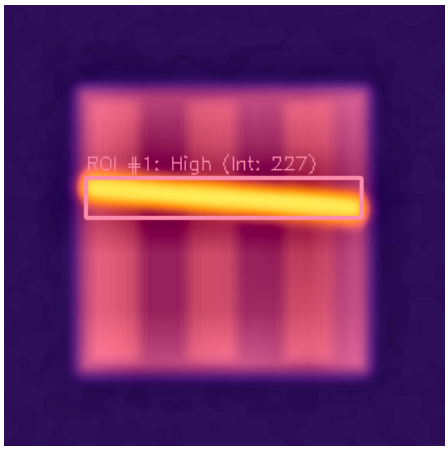
University Engineering Checkpoint • Task–4 Diagnostic Inspection Report

Analysis ID:	3966a945	**Inspection Date:**	2026-11-01 09:54:46 IST
Target Filename:	sample_insulation_fault_1.png	**Model Architecture:**	v1.0-ONNX
Analysis Status:	SUCCESS	**Diagnostic Condition:**	Fault Detected
Classified Fault:	Electrical Fault	**Model Confidence:**	51.80%
Assessed Severity:	High	**Resolution (Orig/Proc):**	320 x 320 px / 224 x 224 px

THERMAL SENSOR IMAGERY & ROI LOCALIZATION



Original Thermal Acquisition



OpenCV Processed & Hotspot ROI

ANOMALOUS THERMAL HOTSPOT COORDINATES

Region	Bounding Box (X,Y,W,H)	Peak Intensity	Area (Pixels)	Local Severity
Primary Hotspot	[59, 125, 200, 29]	227.0	2429 px	High

LIMITATIONS & ACADEMIC PROJECT DISCLAIMER

This report is generated automatically by the Deep Learning-Based Thermal Fault Detection system (University Project Checkpoint Task–4). Thermal analysis is performed through computer vision (OpenCV Java) and deep neural network classification (ONNX Runtime Java). Standard 8-bit RGB colorized infrared imagery does not represent certified absolute thermometric calibration. Results should be validated by certified thermographers before physical maintenance interventions.